

Medical Image Processing
M2 Data Science & AI 2024-2025

www.sop.inria.fr/teams/asclepios/cours/MS_DS/

X. Pennec

Introduction to medical image acquisition



Epione team
2004, route des Lucioles B.P. 93
06902 Sophia Antipolis Cedex
<http://www.sop.inria.fr/epione>

Course teachers

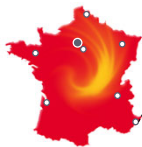
Hervé Delingette



Xavier Pennec



Inria Research Centers



Epione Research Team

1. Biomedical Image/Data Analysis, Machine Learning
2. Imaging & Phenomics, Biostatistics
3. Computational Anatomy, Geometric Statistics
4. Computational Physiology & Image-Guided Therapy
5. Computational Cardiology & Image-Based Interventions

01/10/2024

Medical Image Analysis – M2 DSAI 2024-2025

Thursday afternoon

Course notes : http://www.sop.inria.fr/teams/asclepios/cours/MS_DS/

- Thur. Jan. 09, 14:00-17:00 [Xavier Pennec] Introduction to Medical Image Acquisition
- Thur. Jan. 16 13:30-16:30 [Xavier Pennec] Medical Image Registration
- Thur. Jan. 23 13:30-16:30 [Hervé Delingette] Image Filtering
- Thur. Jan. 30 13:30-16:30 [Xavier Pennec] Statistics on Riemannian manifolds and Lie groups
- Thur. Feb. 6 13:30-16:30 [Hervé Delingette] Principles and Metrics for Image segmentation
- Thur. Feb. 13 13:30-16:30 [Hervé Delingette] Image segmentation based on clustering
- Thur. Feb. 27 13:30-16:30 [Xavier Pennec] Diffeomorphic Registration and Computational Anatomy
- Thur. March 6, 13:30-16:30 [Hervé Delingette] Connectivity and Shape Constrained Image segmentation
- Thur. March 13 2023 13:30-15:00, [XP & HD] EXAM

01/10/2024

Xavier Pennec

3

Medical Image Processing – M2 DSAI 2024-2025

Exam: 4 components :

- Scientific Article Study :
 - [1] 10 min oral presentation
 - [2] 10 min Questions & Answers
 - [3] 5-6 page report presenting the paper and putting it in perspective.
 - Implementation (optional)
- [4] Multiple choice Quizz : 10-15 questions

01/10/2024

Xavier Pennec

4

Course overview

The data of Computational Anatomy & Physiology

Tomography

MRI

Nuclear Medicine

Other modalities

Discrete and continuous images

01/10/2024

Xavier Pennec

5

1895: invention of X-ray imaging



Roentgen



First Nobel prize
in Physics in 1901

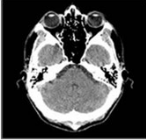
01/10/2024

Xavier Pennec

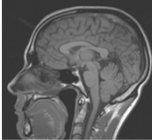
6

Today's Medical Imaging modalities

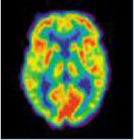
CT Scan



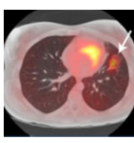
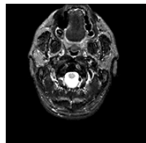
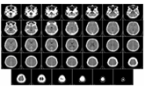
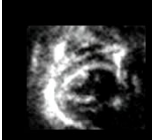
MRI



PET



Ultrasound



Source: IT-Picture

01/10/2024

Xavier Pennec

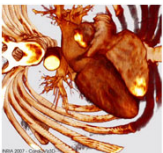
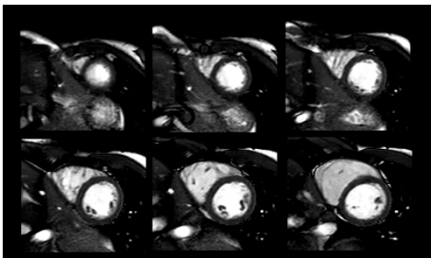
7

Dynamic Images (4-D)

CT Scan



MRI



01/10/2024

Xavier Pennec

8

Bio-signals

ECG



Pressure Sensor



Temperature

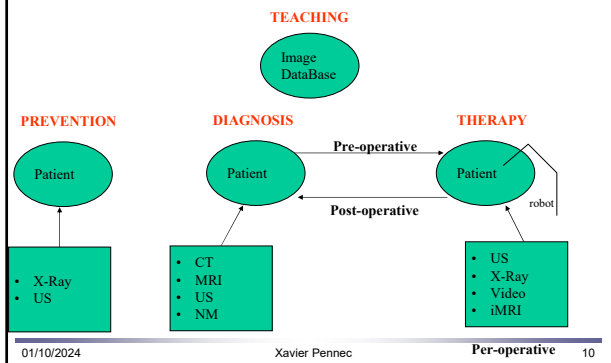


01/10/2024

Xavier Pennec

9

Medical Imaging in clinical practice



Course overview

The data of Computational Anatomy & Physiology

Image acquisition

- Introduction
- Tomography
- MRI
- Nuclear medicine

Medical Image Registration

01/10/2024 Xavier Pennec 11

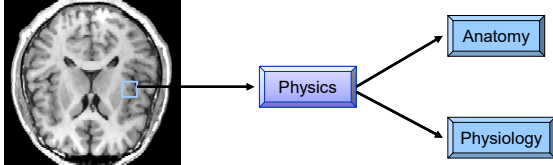
Different imaging modalities

- X-ray
- Magnetic resonance imaging
 - anatomic, functional, angiographic, diffusion, spectroscopic, tagged
- Transmission Tomography (X Scan)
- Nuclear Medicine :
 - Positron emission tomography (PET)
 - Single photon emission tomography (SPECT)
- Ultrasonography
- Histological Imaging, confocal in-vivo microscopy, molecular imaging,...

01/10/2024 Xavier Pennec 12

Characteristics of medical images

Intensity values are related to physical tissue characteristics which in turn may relate to a physiological phenomenon



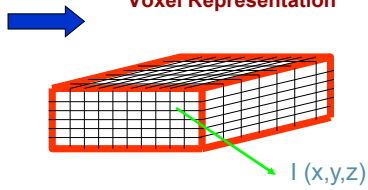
01/10/2024

Xavier Pennec

13

Volumetric medical images

Very often medical images are volumetric
Voxel Representation

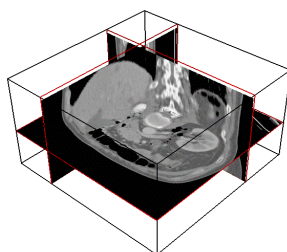
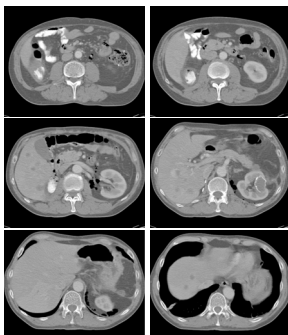


01/10/2024

Xavier Pennec

14

Example of volumetric images : CT-scan (Scanner)



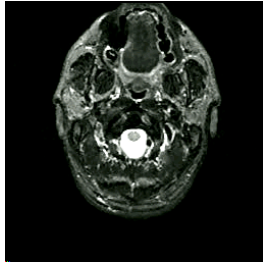
Size: 512 x 512 x 128
Resolution: 0.5 x 0.5 x 1 mm

01/10/2024

Xavier Pennec

15

Volumetric images



T2 MRI

01/10/2024

Xavier Pennec

16

Course overview

The data of Computational Anatomy & Physiology

Tomography

MRI

Nuclear Medicine

Other modalities

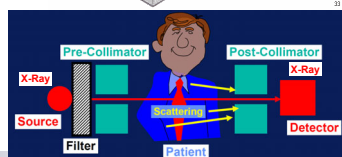
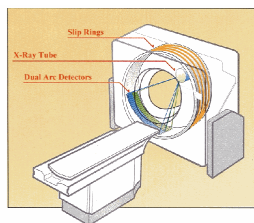
Discrete and continuous images

01/10/2024

Xavier Pennec

17

Principle of CT Imaging (1)



01/10/2024

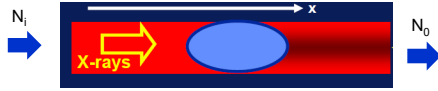
Xavier Pennec

18

Principle of CT Imaging (2)

Input X-Ray intensity : N_i

Measured Output X-Ray intensity : N_o



Exponential attenuation :

$$N_o = N_i e^{-\int_{-\infty}^{+\infty} \mu(x) dx}$$

Objective :

- measure $\mu(x)$ = absorption coefficient of X-ray

01/10/2024

Xavier Pennec

19

Computed Tomography

Principle :

- Reconstruct n dimensional function (image) from projected data of $(n-1)$ dimension

Radon Transform (1917)

- "Two dimension and three dimension object can be reconstructed from the infinite set of projection data".

01/10/2024

Xavier Pennec

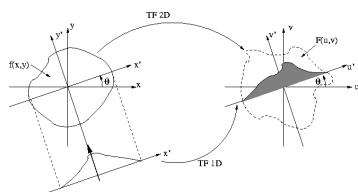
20

Fourier Slice Theorem

1D Fourier Transform of Projected slice of 2D field $\mu(x,y)$

is equal to

1D slice in the same direction of the 2D Fourier Transform of $\mu(x,y)$



01/10/2024

Xavier Pennec

21

Basic principle of CT

-Reconstruction of 2 dimensional image-

Projection Data

curvilinear integral of absorption coefficient regarding Y

The diagram illustrates the basic principle of CT. On the left, the 'Data Acquisition field' shows an X-ray tube (red) and an X-ray detector array (blue) rotating around a central 'object' (blue circle). Multiple projection rays (grey lines) pass through the object. On the right, the 'Reconstruction field' shows the same setup but with the object reconstructed as a series of overlapping blue squares, representing the backprojected data.

Simple Backprojection

01/10/2024

Xavier Pennec

22

Reconstruction Principle

Backprojection based on inverse Radon Transform

$$\overline{\mu(x)} = \frac{1}{2\pi} \int_{\alpha=0}^{2\pi} R\mu(\alpha, n_{\alpha}, x) d\alpha$$

$$n_{\alpha} = (\cos \alpha, \sin \alpha)$$

In practice use filtered back-projection to remove blur

The image shows three square plots side-by-side. The first, 'Model Image', shows a sharp white circle on a black background. The second, 'Simple Backprojection', shows a very blurry white circle. The third, 'Filtered Backprojection', shows a sharp white circle, similar to the model image.

Model Image Simple Backprojection Filtered Backprojection

01/10/2024

Xavier Pennec

23

Spiral (3D) CT

X-ray tube and detectors rotate 360 deg

Patient table is continuously moving

Produce an helix of image projections

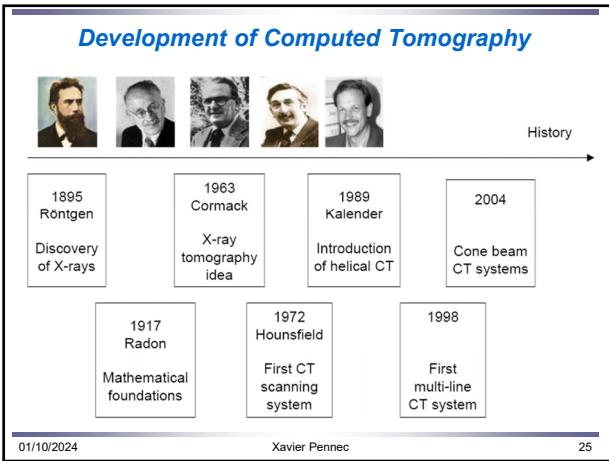
3D reconstruction

The diagram shows a patient lying on a table that moves to the left, indicated by an arrow and the text 'Direction of patient transport'. Above the patient, a helical path is shown, representing the 'Path of rotation tube and detector'. Below the patient, two horizontal bars represent the 'z, mm' and 't, s' axes, with arrows indicating the direction of movement.

01/10/2024

Xavier Pennec

24



CT Scan Imaging

Measure absorption coefficient of X-ray related to density of tissue

Invasive image modality (ionizing rays)

Absolute Hounsfield Unit

$$HU(\mu) = 1024 \frac{\mu - \mu_{\text{water}}}{\mu_{\text{water}}}$$

- HU(water)=0, HU(air)= -1024, HU(Bone)=175 to 3000
- Coded on signed 12 bits

01/10/2024 Xavier Pennec 26

Tomotensitometrie (Scanner X)

01/10/2024 Xavier Pennec 27

Course overview

The data of Computational Anatomy & Physiology

Tomography

MRI

Nuclear Medicine

Other modalities

Discrete and continuous images

01/10/2024

Xavier Pennec

28

Magnetic resonance imaging



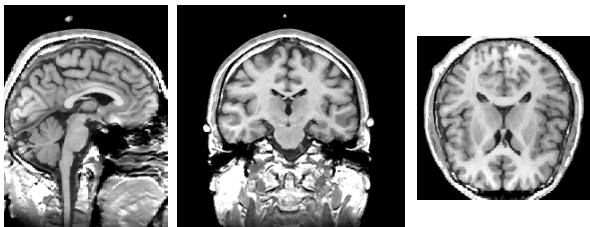
Density and structure of protons

01/10/2024

Xavier Pennec

29

Magnetic resonance imaging



Sagittal

Coronal or Frontal

Axial or Transverse

dimension: 256 x 256 x 128

résolution: 1x1x1.5 mm

01/10/2024

Xavier Pennec

30

MRI: a few dates

- 1946: MR phenomenon - Bloch et Purcell
- 1952: Nobel prize - Bloch et Purcell
- 1950-1970: development but no imaging
- 1980: MRI feasibility
- 1986 - ...: real development

01/10/2024

Xavier Pennec

31

MRI: One modality with multiple sequences

- Anatomic MRI: T1, T2, DP weighted images
- Angiographic MR
- Functional MR: cognitive studies
- Diffusion MR: brain connectivity
- MR Spectroscopy

No absolute quantification

01/10/2024

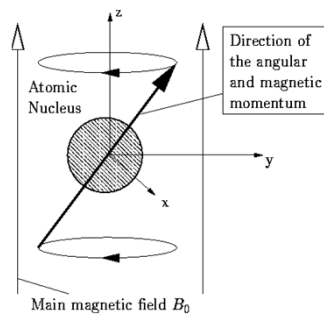
Xavier Pennec

32

Magnetism at the molecular level

Electric charges in motion

- magnetic momentum
- Precession motion in a magnetic field



01/10/2024

Xavier Pennec

33

Bloch's Equations

- Link between spin and magnetic momentum

$$\mu = \gamma S$$

μ Nuclear magnetic momentum
 γ Gyromagnetic ratio

- Fundamental motion equation

$$\frac{dS}{dt} = m$$

S: Angular speed (spin)
m: Momentum (mechanic)

- In a magnetic field $m = \mu \wedge B$

01/10/2024

Xavier Pennec

34

Bloch's Equations

Thus $\frac{d\mu}{dt} = \gamma(\mu \wedge B)$

$$B_0 = \begin{pmatrix} 0 \\ 0 \\ B_0 \end{pmatrix} \rightarrow \mu = \begin{pmatrix} \mu_t \cos(\omega_L t + \varphi) \\ -\mu_t \sin(\omega_L t + \varphi) \\ \mu_{z_0} \end{pmatrix}$$

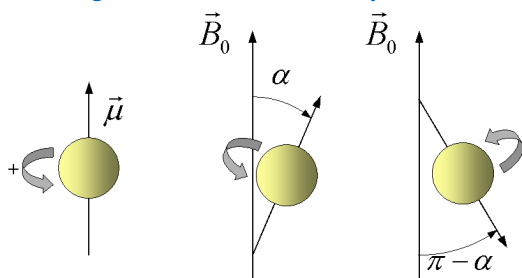
Larmor's frequency $\omega_L = \gamma B_0$

01/10/2024

Xavier Pennec

35

Magnetism at the macroscopic level



$$P_{+1/2} = 0.5000049$$

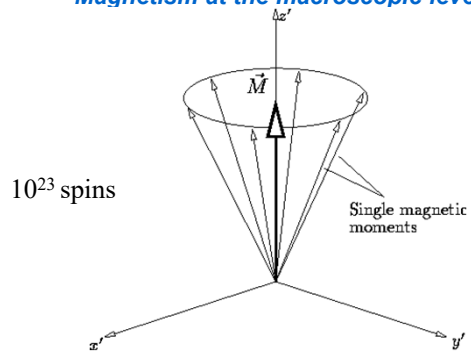
$$B_0 = 1.5 \text{ T}$$

$$P_{-1/2} = 0.4999951$$

01/10/2024

Xavier Pennec

Magnetism at the macroscopic level

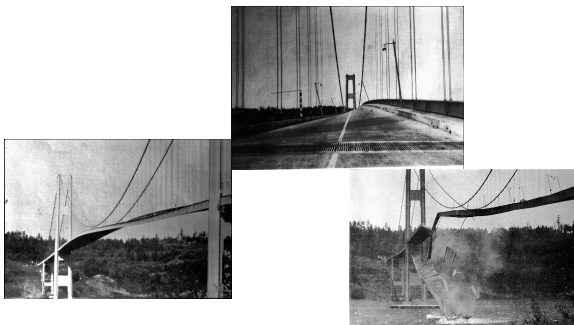


01/10/2024

Xavier Pennec

37

Resonance



01/10/2024

Xavier Pennec

38

Magnetic Resonance / excitation

- Electro-magnetic field at Larmor's frequency

$$\omega_L = \gamma B_0$$

- Hydrogen protons enter into resonance

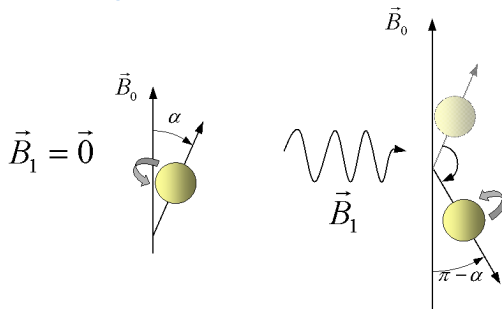
Flip of the macroscopic momentum M

01/10/2024

Xavier Pennec

39

Magnetic Resonance / excitation



01/10/2024

Xavier Pennec

Magnetic Resonance / relaxation

- Return to equilibrium / B_0 : time constant T_1

$$\frac{dM_z}{dt} = \gamma(M \times B)_z - \frac{M_z}{T_1}$$

- Spin dephasing: Time constant T_2

$$\frac{dM_{x,y}}{dt} = \gamma(M \times B)_{x,y} - \frac{M_{x,y}}{T_2}$$

01/10/2024

Xavier Pennec

41

Magnetic Resonance / relaxation

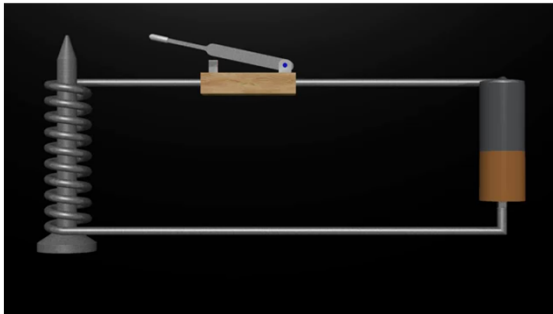
TISSUE	T1 (ms)		T2(ms)
	0.5 T	1.5 T	
Muscle	550	870	45
Heart	580	865	55
Liver	325	490	50
Kidney	495	650	60
Spleen	495	650	58
Fat	215	262	85
Brain, grey matter	655	920	100
Brain, white matter	540	785	90

01/10/2024

Xavier Pennec

42

MRI basics in video



https://www.youtube.com/watch?v=djAxjtN_7VE

01/10/2024

Xavier Pennec

43

MRI / frequency selection

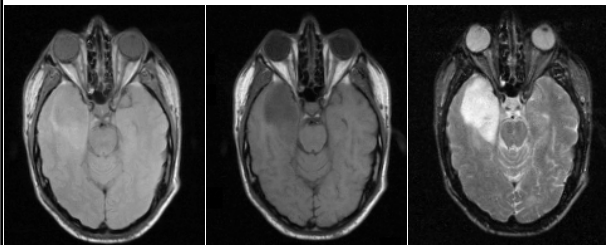
- **X encoding by frequency**
- **Y encoding by phase**
- **Several measures are necessary**

01/10/2024

Xavier Pennec

44

Anatomical MRI



Proton density

T_1

T_2

01/10/2024

Xavier Pennec

45

Angiographic MRI



- INFLOW
- Phases

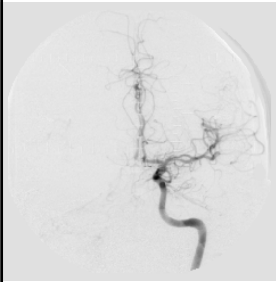
Maximum intensity projection (MIP)

01/10/2024

Xavier Pennec

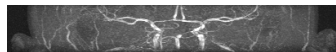
46

Angiographic MRI



X-Scan / radiology

Selective injection of
a contrast agent in
one artery

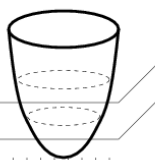


01/10/2024

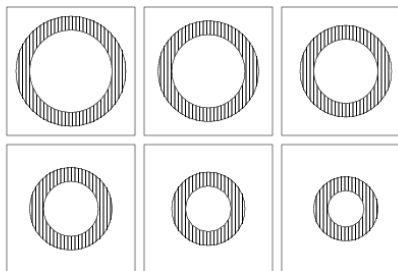
Xavier Pennec

47

Tagged MRI



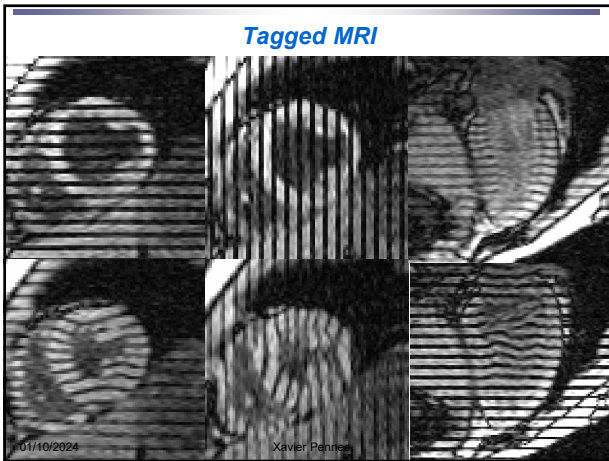
Plans de marquage



01/10/2024

Xavier Pennec

48



Functional MRI

- BOLD : blood oxygen level dependent
 - (oxy)-hemoglobin: diamagnetic
 - deoxy-hemoglobin: paramagnetic

Neuronal activation

- Local oxygen consumption
- Change the ratio of concentration
- Variation of the measured signal

01/10/2024 Xavier Pennec 50

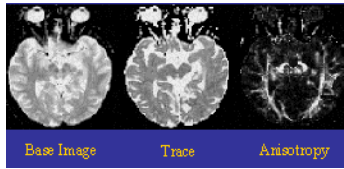
Functional MRI

- Variation of 1-5% for $B_0 = 1.5 \text{ T}$ ($\sim$ noise)
- Variation of 5-20% for $B_0 = 4.0 \text{ T}$

- Statistical analysis on large series
- Motion and intensity biases correction...

01/10/2024 Xavier Pennec 51

Tensor diffusion MRI



01/10/2024

Xavier Pennec

52

Advantages of MRI

- **Non-Invasive:** MRI does not depend on potentially harmful ionizing radiation, as do standard x-ray and CT scans.
- MRI scans are not obstructed by bone, gas, or body waste, which can hinder other imaging techniques
- Can see through bone (the skull) and deliver high quality pictures of the brain's delicate soft tissue structures
- Images of organs and soft tissues

01/10/2024

Xavier Pennec

53

Drawbacks of MRI

- Pacemakers not allowed
- Not suitable for claustrophobic persons
- Tremendous amount of noise during a scan
- MRI scans require patients to hold very still for extended periods of time. MRI exams can range in length from 20 minutes to 90 minutes or more.
- Orthopedic hardware (screws, plates, artificial joints) in the area of a scan can cause severe artifacts
- High cost

01/10/2024

Xavier Pennec

54

Course overview

The data of Computational Anatomy & Physiology

Tomography

MRI

Nuclear Medicine

Other modalities

Discrete and continuous images

01/10/2024

Xavier Pennec

55

Principle of nuclear imaging

- Introduction into the patient body of a couple
(radio-isotope / vector molecule)

Vector Molecule $\Rightarrow$ Targets organ (drug,
protein, blood cells...)

Radio-isotope $\Rightarrow$ Detection of the molecule

- **Emission imaging : the targeted organ emits radioactivity**
 - $\Rightarrow$ Reflect the metabolic function of the organ
 - $\Rightarrow$ *Metabolic or functional imaging*
 - $\rightarrow$ Local relative concentration (relative)
 - $\rightarrow$ Concentration evolution during time

01/10/2024

Xavier Pennec

56

Nuclear Medicine / radioactivity

- **Nucleus (Rutherford)**

A = nucleon number Isobars A = constant

Z = proton number **Isotopes** Z = constant

N = neutron number Isotones N = constant

- **Radioactivity (Curie)**

Alpha: Helium nucleus

Beta: 1/ electron β^-
2/ positron β^+ $\rightarrow$ 2 photons γ (511 keV)

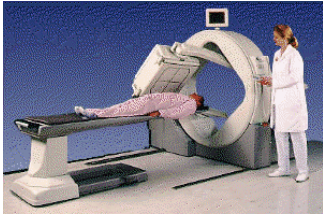
Gamma: Photon

01/10/2024

Xavier Pennec

57

Nuclear Medicine



Density of radioactive tracers

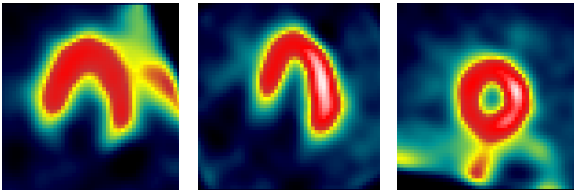
Single photon gamma imaging

Radio-isotopes

Single photon emitters			
Technetium Tc 99m	6 h	140 kev	Portative generator
Iodine I 131	8 j	360 kev	Reacteur (fission)
Iodine I 123	13 h	159 kev	Cyclotron (industry)
Thallium Tl 201	73 h	80 kev	Cyclotron (industry)

Krypton (Kr 81 m), Gallium (Ga 67), Indium (In 111), Xenon (Xe 133, gaz)

Single photon gamma imaging



Heart (myocardium perfusion)

Stress/rest exam

perfusion	/ perfusion	⇒ Healthy area
perfusion	/ (hypo/non-)perfusion	⇒ Zone at risk (ischemia)
(hypo/non-)perfusion	/ (hypo/non-)perfusion	⇒ Infarcted Area

Positron emission tomography (PET)

- Radio-isotopes

Emission : positron (β^+) → Annihilation
→ 2 photons of 511 keV at 180°

Positron emitters

Carbon ^{11}C	20 mn	cyclotron (medical)
Nitrogen ^{13}N	10 mn	cyclotron (medical)
Oxygen ^{15}O	2 mn	cyclotron (medical)
Fluor ^{18}F	112 mn	cyclotron (medical)

- Physiological molecules

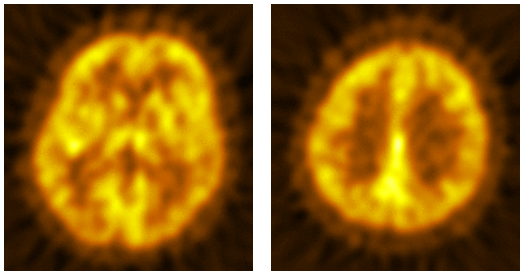
water → H_2O^{15}
glucose → fluoro-deoxyglucose (F^{18}DG)

01/10/2024

Xavier Pennec

61

Positron emission tomography (PET)



01/10/2024

Xavier Pennec

62

PET Scan



<https://www.youtube.com/watch?v=GHLBcCv4rqk>

01/10/2024

Xavier Pennec

63

Course overview

The data of Computational Anatomy & Physiology

Tomography

MRI

Nuclear Medicine

Other modalities

Discrete and continuous images

01/10/2024

Xavier Pennec

64

Echography



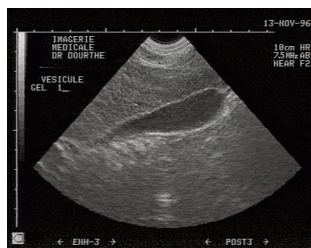
Local variation of acoustic impedance

01/10/2024

Xavier Pennec

65

Echography



Gall Blader

01/10/2024

Xavier Pennec

66

New images

Optical Coherent Tomography (OCT)
Elastometry (MRI, US, etc.)
Spectroscopic Imaging
Terahertz Imaging
Fibred Confocal Imagery *in vivo*
etc.

Source : Mauna Kea Technologies

01/10/2024 Xavier Pennec 67

In Vivo Cellular & Molecular Imaging

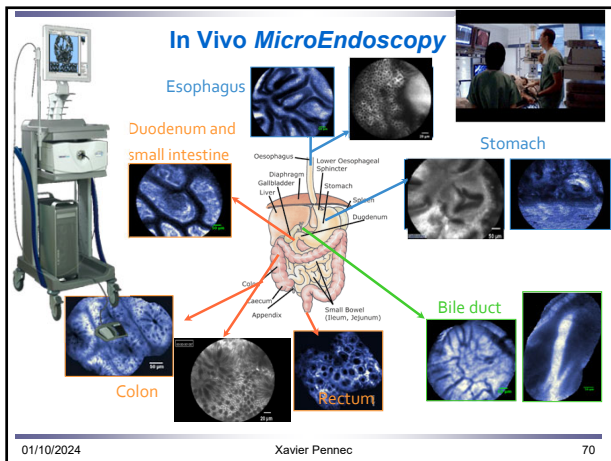
In vivo, in situ fibred confocal microscopy

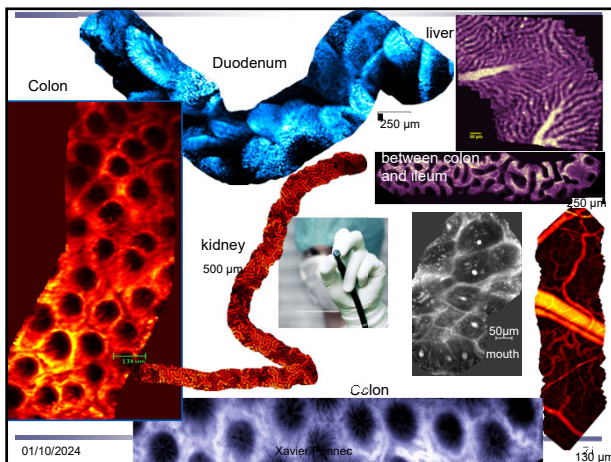
CellVizio, Mauna Kea Technologies, Paris

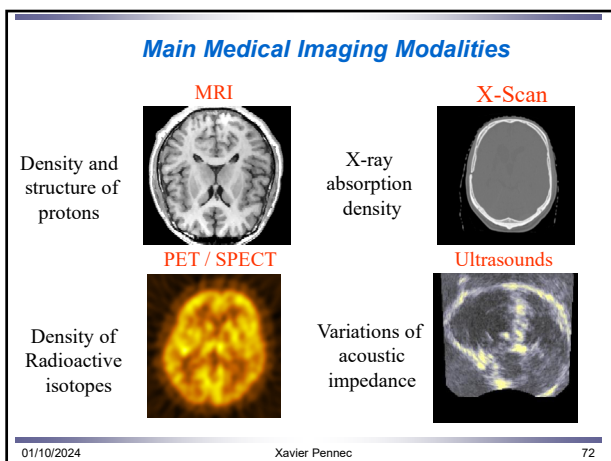
01/10/2024 Xavier Pennec 68

Figure 19.6: Different types of images from Leica FCM1000 microscope (Cellvizio® technology for small animal distributed by Leica microsystems).

01/10/2024 Xavier Pennec 69







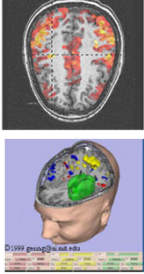
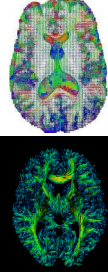
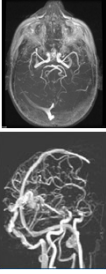
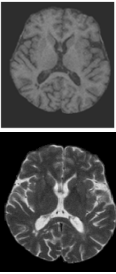
Multiparametric Images

MRI T1, T2

Angio MRI

DTI

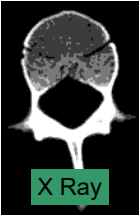
fMRI



01/10/2024Xavier Pennec73

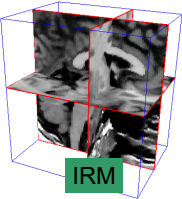
Medical Imaging Classification (1)

Dimensionality



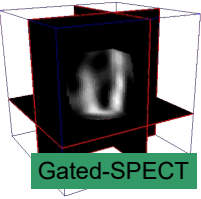
X Ray

2D



IRM

3D



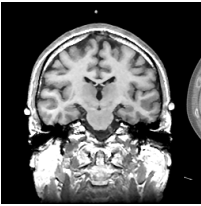
Gated-SPECT

4D (3D+T)

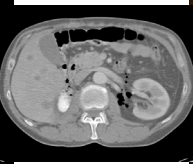
01/10/2024Xavier Pennec74

Medical Imaging Classification (2)

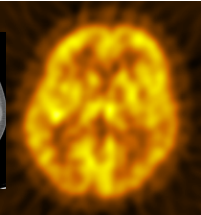
Anatomical vs functional Imagery



MRI



CT with
contrast agent



PET scan

01/10/2024Xavier Pennec75

25

Course overview

The data of Computational Anatomy & Physiology

Tomography

MRI

Nuclear Medicine

Other modalities

Discrete and continuous images

01/10/2024

Xavier Pennec

76

Discrete image representation

- Domain is considered as a 2D/3D regular grid

2D Array I

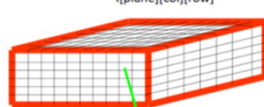
I[col][row]



Pixel=picture element

3D Array I

I[plane][col][row]



Voxel=
Volume Element

01/10/2024

Xavier Pennec

77

Discrete image representation

Pixel / Voxel values can be :

- Discrete :
 - Integer : char (MRI), signed short (CT-scan)
 - Labels of structures
- Continuous :
 - Float / double

Images can be seen as a graph

- Nodes are pixel / voxel centers
- Edges between adjacent elements
- Grid Duality



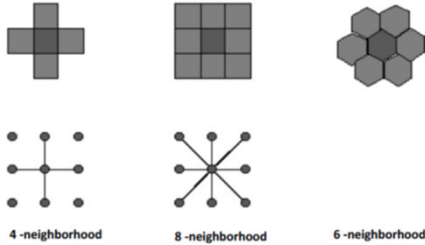
01/10/2024

Xavier Pennec

78

Neighborhoods

Different types of neighborhood in 2D



01/10/2024

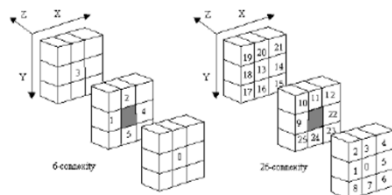
Xavier Pennec

79

Neighborhoods

3 types of neighborhood for a 3D image :

- 6-neighborhood : adjacency through faces
- 18-neighborhood : adjacency through faces and edges
- 26-neighborhood : adjacency through faces and edges and vertices



01/10/2024

Xavier Pennec

80

Continuous image representation

Image seen as 2D or 3D Fields : $I(x)$, $x \in \mathbb{R}^n$, $n = 2, 3$

Requires to define interpolation functions

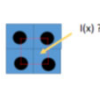


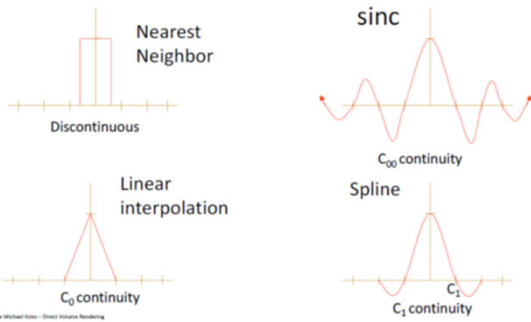
Image Domain	Discrete	Continuous
Image Value	Discrete	Continuous
Discrete	Array of Int	Field of Integer
Continuous	Array of Float	Field of Float

01/10/2024

Xavier Pennec

81

1D interpolation functions



01/10/2024

Xavier Pennec

82

Bilinear interpolation (2D image -> field)

Bilinear Interpolation : 3 linear interpolations



$$I(u, v) = (1-u)(1-v)I_{i,j} + u v I_{i+1,j+1} + (1-u)v I_{i,j+1} + u (1-v)I_{i+1,j}$$

Source : Joe Michael Kiri - Direct Volume Rendering

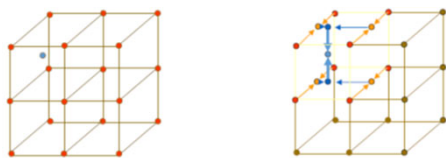
01/10/2024

Xavier Pennec

83

Trilinear interpolation (3D image -> field)

Trilinear Interpolation : 7 linear interpolations



$$I(u, v, w) = (1-u)(1-v)(1-w)I_{i,j,k} + u v w I_{i+1,j+1,k+1} + (1-u)v w I_{i,j+1,k+1} + u (1-v) w I_{i+1,j,k+1} + (1-u)v (1-w)I_{i,j+1,k} + u (1-v)(1-w)I_{i+1,j,k} + u v (1-w) I_{i+1,j+1,k} + (1-u) (1-v) w I_{i,j,k+1}$$

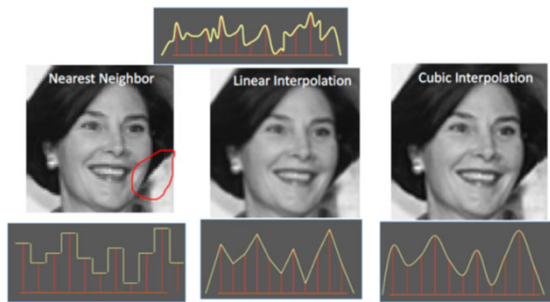
Source : Joe Michael Kiri - Direct Volume Rendering

01/10/2024

Xavier Pennec

84

Image interpolation

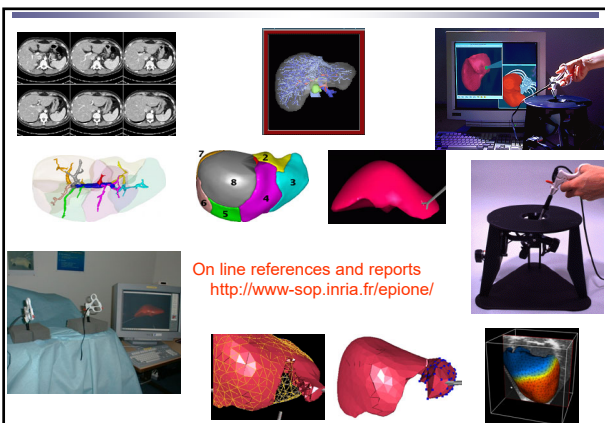


Source : Joe Michael Kriss - Direct Volume Rendering

01/10/2024

Xavier Pennec

85



On line references and reports
<http://www-sop.inria.fr/epione/>

01/10/2024

Xavier Pennec

86
